



DESIGN AND ANALYSIS OF A HUMIDITY CHAMBER

A report submitted as a part of SIRE Internship by

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ABSTRACT

Almost every manufactured product like manufactured products like electrical appliance, batteries, machine tools, Automotive body parts etc. interact with the atmosphere so they need to be tested for their durability and how the natural atmospheric conditions going to affect them. So before getting the product to market its manufacturing flaws and faults are needed to be tested by subjecting them to various environmental. These tests are necessary to ensure product quality and to avoid unnecessary loss due to manufacture of faulty components. Many industries such as food processing, manufacturing, engineering and packaging performs climatic related environmental tests to ensure product reliability and quality.

Humidity chambers which can also be called climate test chambers are mainly used to simulate the thermal environmental conditions and observe the effect of a variety of range of temperature and humidity have on a product or component. The chamber itself consumes lot of energy during its operation, this huge energy consumption serves as an issue of concern to researchers globally

In this report, a humidity chamber is designed and analysed with initial fluid fluent analysis on ANSYS Workbench 18.2 taking temperature, streamline and volume fraction at different temperature.

INTRODUCTION

2.1 Psychrometrics

The physical and thermodynamic properties of gas vapor mixture are known in mainly three names, mainly, Psychrometrics, Psychrometry, Hygrometry. The principle of Psychrometry can be applied to various physical system containing gas vapor mixture, one of the most common research topics are on water vapor and air mixtures, mostly because of its uses in modern day heating, cooling, air conditioning, also meteorology. To sum up, basically thermal comfort depends on the saturation of air with water vapor.

2.2 Psychrometric properties

Dry bulb Temperature (DBT)

It is the temperature indicated by a thermometer exposed to air, the term dry air is added to temperature to distinguish it from wet bulb and dew point temperature. Usually the word temperature by itself without a prefix usually means dry bulb temperature, the name implies that the sensing bulb or element is dry.[1]

Wet bulb temperature (WBT)

The wet bulb temperature is a thermodynamic property of mixture of air and water vapor. The accuracy depends on how fast the air passes over the bulb.

Dew point temperature (DPT)

It is the saturation temperature of moisture present in a sample of air

Humidity

Humidity is the strength of water vapor present in air. Humidity plays a significant role for surface life, for animals and humans depending on moisture to regulate internal body temperature, high humidity levels can reduce the exchange efficiency of rate of moisture evaporation from skin surfaces

Specific humidity

It is defined as mass of water vapor as a quantity of the mass of the moist air sample (including both dry air and water vapor)

Absolute humidity

The ratio of mass of water vapor to unit volume of air containing of the water vapor. It is also known as the water vapor density.

Relative humidity

It is defined as the ratio of the vapor pressure of moisture in the sample to the saturation vapor pressure of the dry bulb temperature of the sample.

Psychrometric ratio

The psychrometry ratio is the ratio defined as the heat transfer coefficient to the product of humid heat and mass transfer coefficient at a dampened surface. The following equation gives the value,

$$r = \frac{h_c}{k_y c_s}$$

where

r = Psychrometric ratio, dimensionless h_c = convective heat transfer coefficient, $\text{Wm}^{-2}\text{K}^{-1}$ k_y = convective mass transfer coefficient, $\text{kgm}^{-2}\text{s}^{-1}$ c_s = humid heat, $\text{Jkg}^{-1}\text{K}^{-1}$

+

2.3 Psychrometric charts

Terminology

A psychrometric chart is a graph of the thermodynamic properties of moist air at a pressure (constant), often equated to an elevation relative to sea level [2].

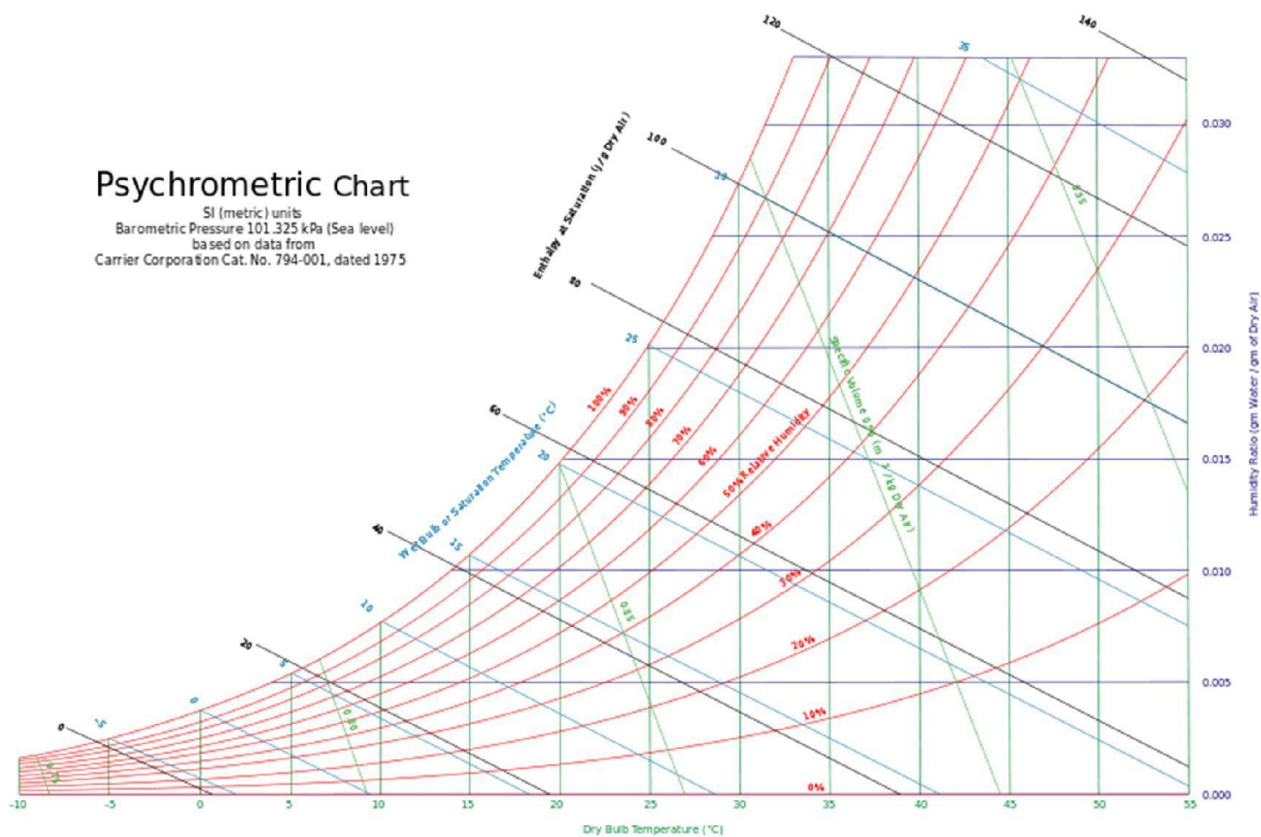


Fig 1. Psychrometric Chart for Calculations [2]

2.5 Humidity Chamber

Humidity Chamber is a dynamic testing apparatus for evaluating the prolonged effect of humidity on various manufactured components to ascertain its quality parameters in such extreme conditions. This instrument is not subjected to only one industry applications but used in various pharmaceuticals, plastic and rubber industries for quality assurance testing [3].

Humidity chambers also known as Climate chambers, there are roughly two types of climate chambers, constant and dynamic climate chambers. Constant climate chambers are mainly used for long term stability tests and can be used to preserve temperature and humidity at definite levels. On the other hand, dynamic climate chamber can be used to run specific test programs in which temperature and humidity are modified automatically at specific intervals.

Constant climate chambers are used in industrial and research fields, some applications include,

Rubber Industries – Rubber Industries use them to govern the initiation of crack and also melting of rubbers in environment (hot and cold).

Pharmaceutical Industries – Pharmaceutical Industries also test the effect of highly humid environment on medicines and drugs to evaluate extreme conditions.

Plastic Industries – Plastic Industries use Climate chamber to determine the effect on several mechanical and physical properties of various plastic products in extreme environment and high temperatures."

Atomizers

They consist of a fine spray nozzle that help in atomizing the water. This is done in an assembly called the mixing assembly. As this atomized water passes through the heater in the chamber, temperature instantly increases and it heats up. Hence, atomizers are commonly used in humidity chambers to generate a high level of humidity according to specification and offer abundant benefits. Some of them are cited below:

1. Atomizers are highly capable of generating high volume of moisture according to specifications
2. They create no issues with heater burn and function well.
3. They are simple and easy to use systems
4. They save electricity while giving maximum output
5. Also really efficient cooling the air at good efficiency

Humidity and Temperature tests

These tests are performed by keeping the products to be tested in artificial environments with specific temperature and humidity levels. The testing chamber size varies based on the components to be tested, small ones are used to test miniature components like circuit boards and controllers, mid-sized are mainly used for testing small products for mass testing or to allow much larger components to be tested. Sometimes, vehicles and machines that need to be tested are tested in custom built chambers.

One of the most common temperature and humidity testing includes 85/85 test also known as the damp heat test, this test has a testing threshold that is used in many tests. It involves setting the chamber condition to 85 degrees Celsius and a humidity value at 85%, these conditions are held as long as 1000 hours.

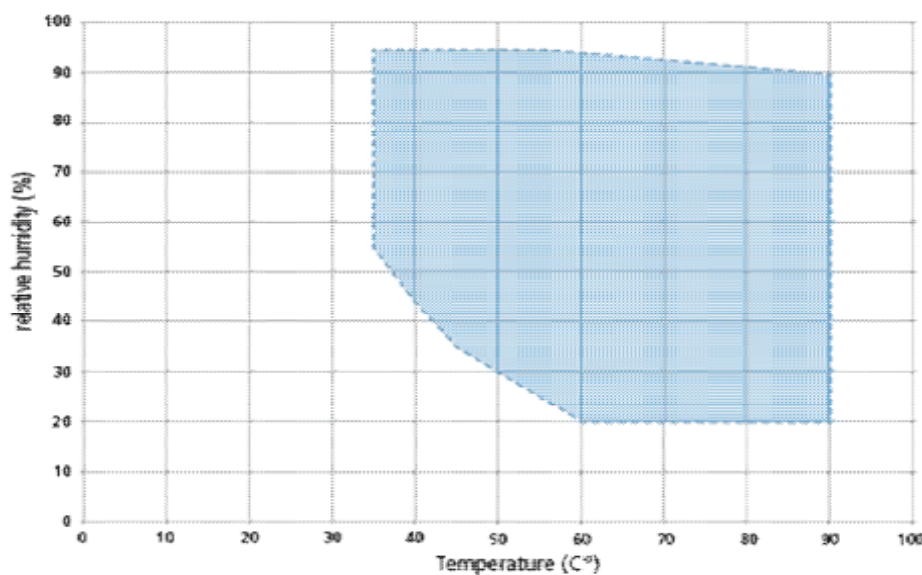
Although a 1000 hours may seem long, it very possible in the operational world where many products are expected to perform.

To reduce the test timing, experts use two processes

High Accelerated Life Testing (HALT)

It involves aggressively using temperature and humidity with multi axis shock vibrators for inducing real time stress.

HALT test find out the product weaknesses and strength flaws much faster than passive chambers. This helps in faster detection of flaws in prototype stages and enables engineers to quickly react with solutions



Temperature-humidity working range HCP

Fig 2 Temperature VS Humidity graph

Highly Accelerated Stress Screening (HASS)

HASS is mainly used for already quality test passed final products that need a final quality control check before it goes out for distribution and manufacture. HASS is less aggressive compared to HALT

Another term often used mainly with humidity and temperature testing, it involves subjecting the product to a controlled cycle of hot/cold and wet/dry conditions. Another test involved is thermal shock testing which involves heating or cooling a product to its maximum range and then quickly reversing the temperature, since most products don't need to undergo such extreme quality testing, these are mostly done on military grade and aviation industry components

Some of the weaknesses identified with these tests are.

Mechanical failure:

These changes in humidity and temperature can lead to a mechanical failure in the component

Water tightness:

Mainly water leakages and ingress weakness are exposed from these

Fogging:

Product weaknesses quickly emerge when subjected to hot and cold cycle, damp and dry. Moisture fogging occurs between the layers which is a sign of flaw

LITERATURE REVIEW

Generally, a temperature and humidity chamber is equipped with refrigerator, heater, and humidifier for realizing the thermal environmental condition. The temperature and humidity chamber consumes a lot of energy during its operation. Therefore, the energy consumption rate of conventional temperature and humidity chambers serves as an issue of concern for researchers globally. Most of researches about the temperature and humidity chambers have focused on increasing the accuracy of the temperature and humidity by the electrical control method. Sami and Partenen [3] conducted experimental research on to evaluate the performance of an environmental testing chamber and the aim of this thesis study mainly was to get all the vital information from the BGD 897/408B Climate testing chamber by doing different types of unique tests. this information can be used for refining test cycles to be as fast as possible, taking into account the working principle of the chamber, all the errors, and gathering proper results from tests. Abdel-Salem et al., [4] investigated experimentally the influence of inlet air and water temperatures, water to air flow ratio to the humidifier on fresh water production. They observed that an increase system productivity with at high air and water temperatures to the humidifier. Yanniotis and Xerodemas [5] performed experimental and computational fluid dynamics (CFD) studies on spray and pad humidifiers. Comparative experimental and CFD results gives better evaporation with pad humidifier with 300 mm pad thickness for high air to water flow ratios. Detailed design and analysis was carried out by Ettouney [6] on different layouts of humidification dehumidification desalination process and concluded that accumulation of air in a condenser is to be taken care by proper design of the dehumidification unit. Chiranjeevi and Srinivas [7,8] studied the influence of different operating parameters on the performance of an integrated solar desalination plant theoretically and experimentally. They conducted experiments with PVC sheets as packing material in the humidifier and observed

an increased air humidification at high temperatures and high flow rates of hot water and results in increased fresh water yield

Data on temperature and humidity as physical quantities were gathered. Environmental tests were designed considering the quantities that needed to be verified. The key areas important in these tests were the static and dynamic accuracy of temperature and RH control, relation and interaction of these two quantities, and capabilities of the climate chamber. In future, this information obtained from this research study could be used to optimize the test cycles to be as efficient and fast as possible.

OBJECTIVES OF THE REPORT

In the literature review we can see the different tests that are run on the Environmental Testing Chamber /Humidity Chamber/Climate Chamber to get information on temperature and humidity.

The key areas important in these tests were the static and dynamic accuracy of temperature and RH control, relation and interaction of these two quantities, and capabilities of the climate chamber. In future, this information obtained from this research study could be used to optimize the test cycles to be as efficient and fast as possible.

METHODOLOGY

A humidity chamber is designed for analysis and the humidity level in the temperature controlled dry air is controlled.

For humidity control, the cold dry air is passed through multiple atomizers which help add moisture to the dry air according to the requirement. And the water volume fraction at different locations is found out. Fig 3a. represents the humidifying chamber in which water is sprayed using atomizer through the nozzle as shown in Fig.3b.

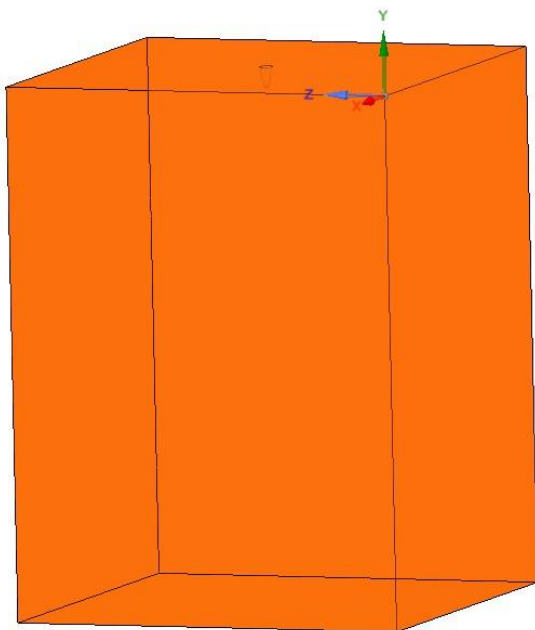


Fig.3a. 3D Model of Humidifier Chamber

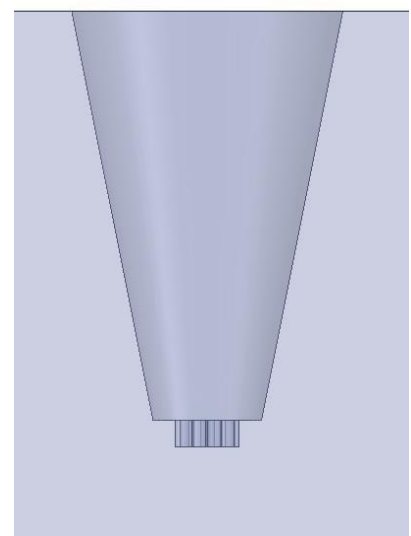


Fig.3b. Nozzle

RESULTS

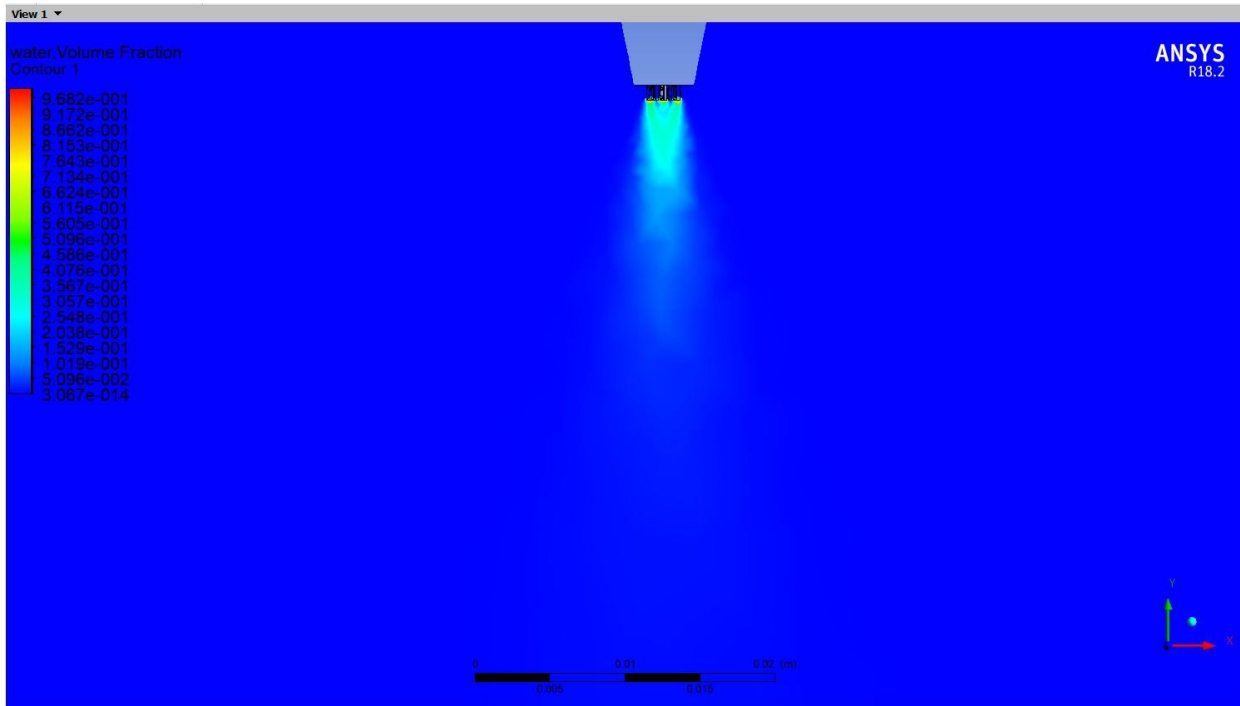


Fig.4a. Volume fraction of water in the chamber at the beginning

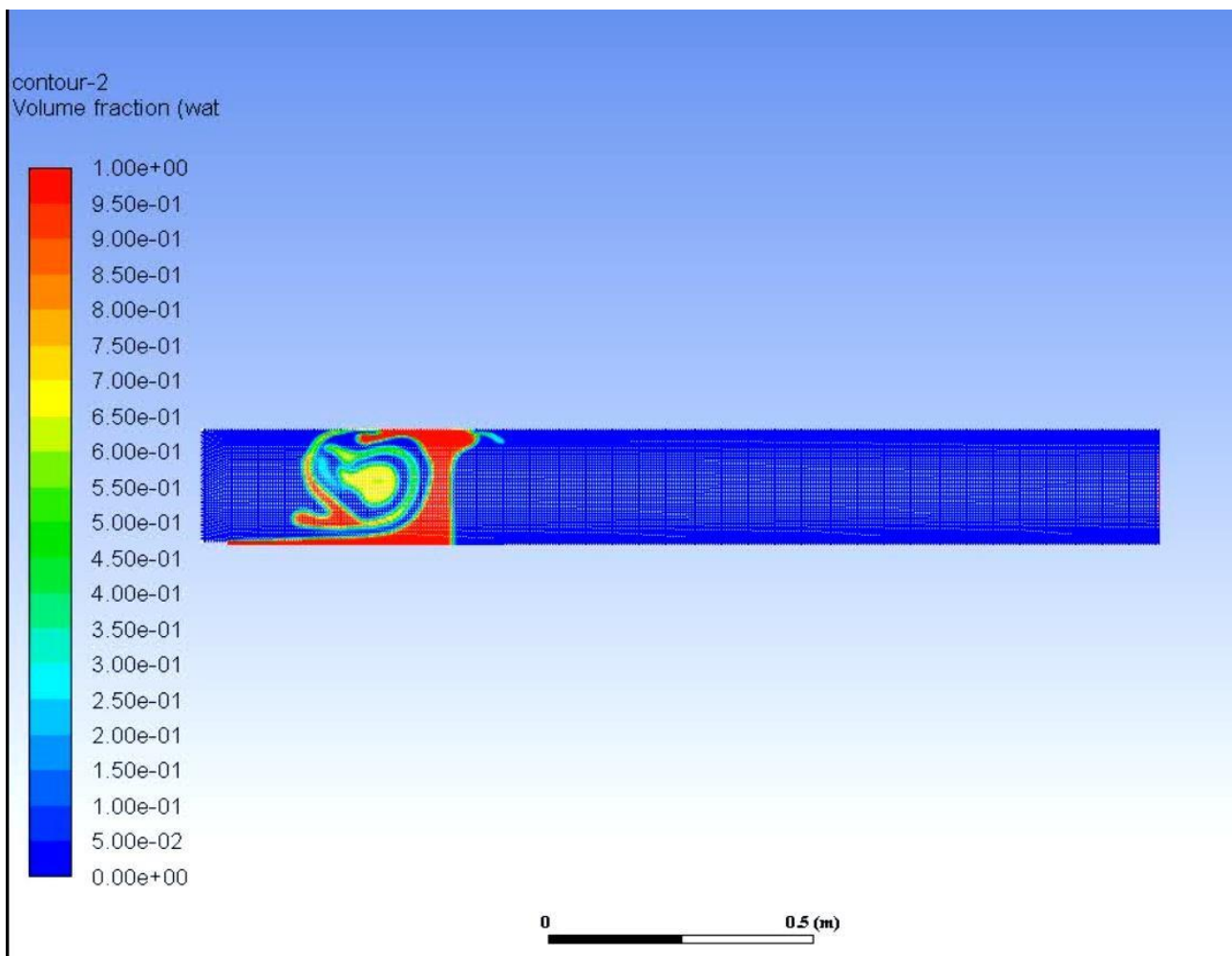


Fig.4b. water volume fraction at the end of 5s in the humidity chamber

Fig. 4a. and 4b. represent the volume fraction of water vapours in the humidity chamber at the beginning and after 5s of spraying.

CONCLUSION

On keeping the input water temperature from the atomizer as 25 degrees Celsius and input. The different parameters on the atomizer chamber was also obtained. Hence the humidity chamber analysis results were obtained and further design changes are to be determined to obtain an efficient humidity chamber that have less process times and better material.

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